

Assignment 2 Report

Building Detection from Satellite Imagery

1. Problem

The objective of this project is to automatically detect and count buildings in satellite images using a deep learning object detection model. Building detection is useful in urban planning, disaster management, population estimation, infrastructure monitoring, and mapping applications. The final system takes a satellite image as input and returns the detected buildings along with their count.

2. Data

The project uses the ICG SpaceNet Sandbox YOLO Dataset available on Kaggle

The dataset contains 1,148 satellite images and 1,148 corresponding YOLO-format label files, resulting in 1,148 correctly paired image-label samples. The dataset was organized into training, validation, and test sets using an 80:10:10 split. The images are generally consistent in quality and size, but challenges such as dense building clusters, small rooftops, and shadows make building detection more difficult.

3. MODEL

- What is YOLOv8?: YOLOv8 (You Only Look Once, version 8) is a state-of-the-art, real-time object detection model built by Ultralytics. It is designed to be highly fast and accurate for identifying objects in images.
- Which size did you pick: YOLOv8s (Small variant) using Transfer Learning.
- How many parameters: The model has 11,125,971 parameters (approx. 11.1 million).

model parameters: GFLOPs : 28.4
Layers : 73

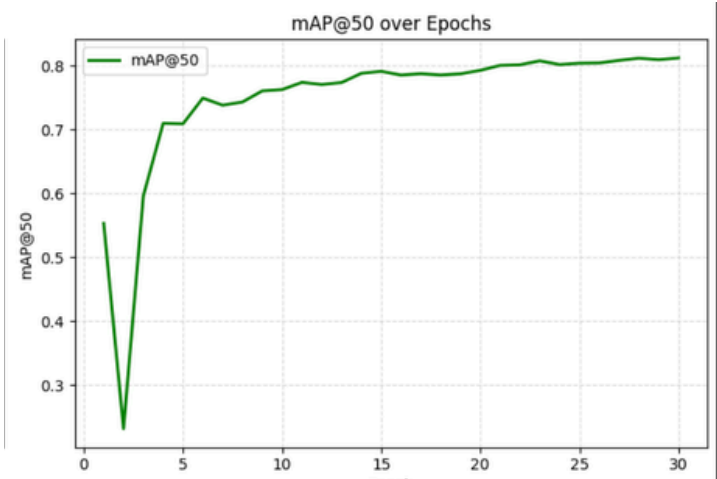
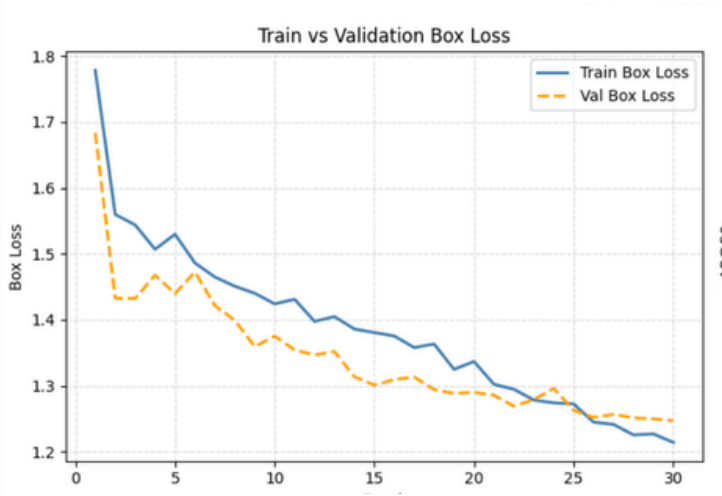
4 . TRAINING

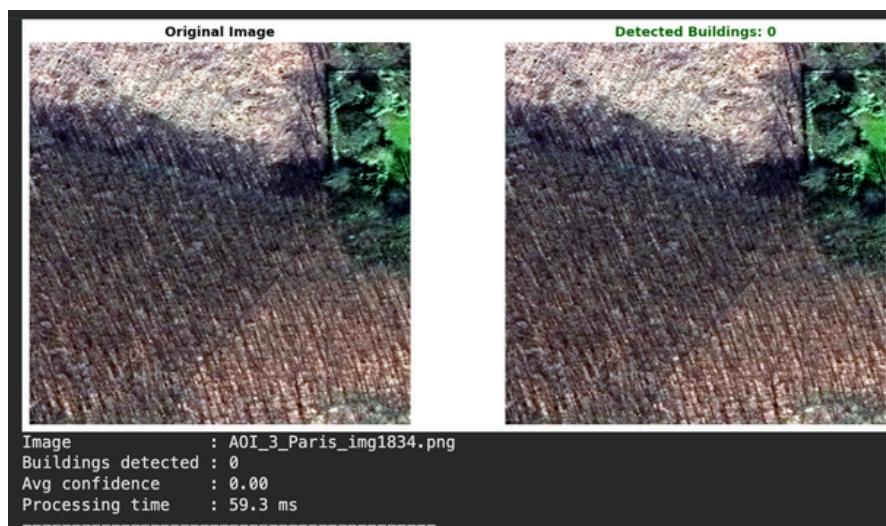
- GPU used: Tesla T4 (Google Colab)
- The YOLOv8s model was trained for 30 epochs with a batch size of 16 and an image size of 640×640. The complete training process took approximately 0.208 hours (12.5 minutes), after which the model achieved an mAP@50 of 0.813 on the validation set.

5. RESULTS

Test Set Performance

Metric	Value	Interpretation
mAP@50	0.8043	About 80% localization accuracy at IoU = 0.5
Precision	0.8328	83.28% of predicted buildings were correct
Recall	0.7163	71.63% of actual buildings were detected
F1 Score	0.7701	Balanced detection performance





6. OBSERVATIONS

What Worked Well

- YOLOv8 successfully detected most buildings across different satellite scenes.
- The model achieved over 80% mAP@50.
- Building counting performance was strong with an MAE of only 2.5.
- Transfer learning using pretrained weights significantly improved training efficiency.

Challenges Faced

- Dense urban areas caused overlapping detections.
- Small buildings were occasionally missed.
- Building shadows sometimes confused the detector.
- Closely packed rooftops could be merged into a single detection.

Possible Improvements

One potential improvement is to train for more epochs using a larger model such as YOLOv8m or YOLOv8l and apply data augmentation techniques. This could improve recall and help detect smaller buildings more accurately.